

Prediction Market Statistics RAGDoll — PDF Validation Handoff

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Branch: master
Status: pushed for operator PDF validation

PURPOSE

This handoff preserves the latest convergence around the Polymarket Agents / prediction-market statistics idea as a research-only Lantern OS / RAGDoll skill seed.

It is intended to support a later human-readable PDF validation pass.

EVIDENCE BOUNDARY

The uploaded Polymarket Agents README evidence showed:

- the upstream repository is archived and read-only;
- the repository describes a Python 3.9 developer framework for Polymarket agents;
- it includes Polymarket API integration, Gamma market metadata tooling, CLOB/order tooling, local/remote RAG support, news/data sourcing, Pydantic objects, and CLI market listing;
- it also contains autonomous trading entry points;
- its Terms of Service note prohibits U.S. persons and people in certain jurisdictions from trading through Polymarket UI/API/agents.

Lantern OS should therefore treat this as research architecture, not a live betting system.

CONVERGED SKILL

```
ragdoll.skill.prediction_market_statistics.v1

Mode:
research-only by default

Inputs:
market metadata
current prices
volume/liquidity
resolution rules
public evidence
related markets
historical/base-rate data

Core outputs:
implied probability
base-rate probability
evidence-adjusted probability
confidence band
ambiguity score
liquidity/slippage risk
legal/compliance gate
recommended action: reject / watch / paper-trade / human-review
```

AGENT TABLE

Agent	Statistical job	Output
Market Scanner	Find liquid markets with enough volume	candidate list
Resolution Agent	Parse settlement criteria	valid / ambiguous / reject
Price Agent	Pull current Yes/No prices	implied probability

Agent	Statistical job	Output
History Agent	Track price movement over time	trend / volatility
Evidence Agent	Gather public sources	source bundle
Base-rate Agent	Compare to historical frequency	prior probability
Arbitrage Agent	Check related outcomes sum near 100%	inconsistency flags
Stance Agent	Summarize comments/news sentiment	directional signal
Risk Agent	Block insider/private/thin/ambiguous markets	allow / reject
Human Operator	Final decision	no action / watch / paper trade / legal trade

RESEARCH TRIAGE TABLE

Market type	Statistical edge to look for	Useful metric	Fleet action
Sports favorites	Mispriced injury/news reaction	price move vs confirmed news	watch / paper trade
Sports coin-flips	Spread disagreement across markets	45-55% drift plus volume	compare only
Elections	Polling/base-rate mismatch	market probability vs polling average	research brief
Crypto targets	Volatility-implied probability mismatch	price distance plus time remaining	alert only
Fed / CPI / rates	Calendar-driven stale markets	expected range vs market odds	high-value research
Awards	Public sentiment vs insider-like buzz risk	nomination history plus guild data	cautious brief
Geopolitics	Usually reject	source quality / insider risk	block
Viral/news	Resolution ambiguity	clarity score	reject unless clear
Same-day markets	Fast public-info lag	timestamped source delta	alert only

HARD GATES

- 1.** Do not trade autonomously.
- 2.** Do not trade from restricted jurisdictions.
- 3.** Do not use private, leaked, employer, confidential, classified, medical, military, or insider information.
- 4.** Do not treat common odds as edge.
- 5.** Do not use prediction markets as the primary \$200-now plan.
- 6.** Use this as a demonstrable analytics/RAG/fleet asset instead.

\$200-NOW PACKAGING NOTE

The strongest monetizable package remains a fixed-scope audit offer:

- \$199 MCP / RAG / GitHub automation audit
- one-page findings report
- optional cleanup PR or issue list
- 24-hour turnaround

The prediction-market statistics skill can be shown as a demo of the operator's fleet capability, not as a promise of trading profit.

PDF VALIDATION CHECKLIST

A validated PDF should be:

- human-readable, not filepath spam;
- clear that this is research-only;

- clear about legal/compliance gates;
- clear about the statistical value of the repo;
- clear that the fastest \$200 path is service packaging, not gambling;
- formatted with concise tables and no unsupported profit claims.